Baseline Maths Assessment

Year 4

This resource corresponds with the New National Curriculum objectives for Year 3, and has been designed to be used as an aide in assessing pupils' baseline mathematical knowledge at the start of Year 4.

The content addresses each curricular objective outlined in the New National Curriculum in chronological order to give insight into strengths or gaps in each child's mathematical knowledge.

More Assessment resources.

Baseline Maths Assessment – Year 4

Year 3 National Curriculum Objectives:

Number – number and place value

1. Count from 0 in multiples of 4, 8, 50 and 100; find 10 or 100 more or less than a given number

2,3. Recognise the place value of each digit in a three-digit number (hundreds, tens, ones)

4. Compare and order numbers up to 1000

5. Read and write numbers up to 1000 in numerals and in words

1, 2, 3, 4 & 5. Solve number problems and practical problems involving these ideas

Number – addition and subtraction

6. Add and subtract numbers mentally, including a three-digit number and ones

7. Add and subtract numbers mentally, including a three-digit number and tens

8. Add and subtract numbers mentally, including a three-digit number and hundreds

9 & 10. Add and subtract numbers with up to three digits, using formal written methods of columnar addition and subtraction

11. Estimate the answer to a calculation and use inverse operations to check answers

12. Solve problems, including missing number problems, using number facts, place value, and more complex addition and subtraction

Number – multiplication and division

13. Recall and use multiplication and division facts for the 3, 4 and 8 multiplication tables

14 & 15. Write and calculate mathematical statements for multiplication and division using the multiplication tables that they know, including for two-digit numbers times one-digit numbers, using mental and progressing to formal written methods

16. Solve problems, including missing number problems, involving multiplication and division, including positive integer scaling problems and correspondence problems in which n objects are connected to m objects

Baseline Maths Assessment – Year 4

Year 3 National Curriculum Objectives:

Number – fractions

17. Count up and down in tenths; recognise that tenths arise from dividing an object into 10 equal parts and in dividing one-digit numbers or quantities by 10

18. Recognise, find and write fractions of a discrete set of objects: unit fractions and non-unit fractions with small denominators

19. Recognise and use fractions as numbers: unit fractions and non-unit fractions with small denominators

20. Recognise and show, using diagrams, equivalent fractions with small denominators

21. Add and subtract fractions with the same denominator within one whole [for example, 5/7 + 1/7 = 6/7]

22 & 23. Compare and order unit fractions, and fractions with the same denominators

24. Solve problems that involve all of the above

Measurement

25. Measure, compare, add and subtract: lengths (m/cm/mm); mass (kg/g); volume/capacity (l/ml)

26. Measure the perimeter of simple 2-D shapes

27. Add and subtract amounts of money to give change, using both £ and p in practical contexts

28. Tell and write the time from an analogue clock, including using Roman numerals from I to XII, and 12-hour and 24-hour clocks

* Estimate and read time with increasing accuracy to the nearest minute; record and compare time in terms of seconds, minutes and hours; use vocabulary such as o'clock, a.m./p.m., morning, afternoon, noon and midnight

29. Know the number of seconds in a minute and the number of days in each month, year and leap year

30. Compare durations of events [for example to calculate the time taken by particular events or tasks]

*this objective is practical so is not included in this assessment

Baseline Maths Assessment – Year 4

Year 3 National Curriculum Objectives:

Geometry – properties of shapes

31 & 32. Draw 2-D shapes and make 3-D shapes using modelling materials; recognise 3-D shapes in different orientations and describe them

31.Recognise angles as a property of shape or a description of a turn

33, 34 & 35. Identify right angles, recognise that two right angles make a half-turn, three make three quarters of a turn and four a complete turn; identify whether angles are greater than or less than a right angle

36. Identify horizontal and vertical lines and pairs of perpendicular and parallel lines

Statistics

37. Interpret and present data using bar charts, pictograms and tables

38. Solve one-step and two-step questions [for example, 'How many more?' and 'How many fewer?'] using information presented in scaled bar charts and pictograms and tables

Baseline Maths Assessment – Year 4

Number and Place Value

1. Complete each sequence of numbers.

0	4	8	12						
0	8	16	24						
0	50	100	150						
0	100	200	300						

2. Find the following:

10 more than 749		100 less than 399		100 more than 825	
100 less than 793		10 more than 305		10 less than 803	

3. Write the value of each underlined digit.

<u>2</u>98	_____	3<u>5</u>3	_____	14<u>6</u>	_____
3<u>3</u>7	_____	<u>5</u>02	_____	84<u>2</u>	_____

4. Use the symbols > or < to compare the numbers.

964		694	525		552	717		720

5. Write the following numbers in words.

426	_____	708	_____
695	_____	837	_____

Addition and Subtraction

6. Solve the following problems mentally.

839	+	7	=		225	−	6	=	
384	−	6	=		199	+	9	=	

7. Solve the following problems mentally.

209	+	30	=		783	−	40	=	
462	−	80	=		591	+	70	=	

8. Solve the following problems mentally.

827	+	100	=		920	−	500	=	
370	−	200	=		638	+	300	=	

9. 839 + 287 =

10. 793 – 467 =

11. Estimate the answer:

529 + ☐ = 936

Now use the inverse calculation to check your answer.

12. Find the missing numbers:

637 + ☐ = 1,083

☐ – 382 = 513

Multiplication and Division

13. Complete the multiplication and division facts for 3, 4 and 8.

3 x 6 =	8 x 9 =	7 x 4 =	36 ÷ 3 =	8 x 5 =
4 x 8 =	32 ÷ 8 =	11 x 8 =	3 x 12 =	24 ÷ 4 =
27 ÷ 3 =	6 x 6 =	36 ÷ 4 =	11 x 4 =	12 x 8 =

14. Teddy pays £2 for an ice cream. How much would 6 ice creams cost?

15. Amir pays £12 for a football shirt. How much would 4 shirts cost?

16. Charlotte wants to buy sweets for her class. There are 32 students in her class and 10 sweets in each bag. How many bags should she buy so each child gets 1 sweet?

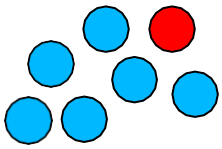
Fractions

17. Complete the sequences.

$\frac{1}{10}$ $\frac{2}{10}$ $\frac{3}{10}$ $\frac{4}{10}$ ☐ ☐ ☐ ☐

$\frac{9}{10}$ $\frac{8}{10}$ $\frac{7}{10}$ $\frac{6}{10}$ ☐ ☐ ☐ ☐

18. Write a fraction to represent the number of red circles and the number of blue circles.



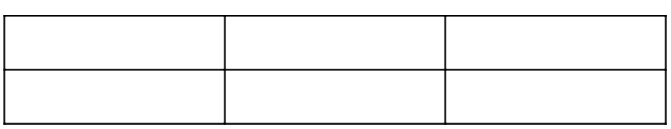
red ☐ blue ☐

19. Calculate the fraction of each number.

$\frac{1}{5}$ of 20 = ☐ $\frac{1}{4}$ of 32 = ☐

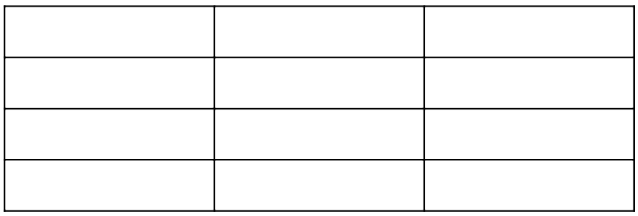
$\frac{2}{3}$ of 9 = ☐ $\frac{2}{5}$ of 35 = ☐

20. Shade the shapes to show three fractions that are equivalent to $\frac{1}{3}$.



☐

☐



☐

21. Complete the calculations.

$\frac{1}{4} + \frac{2}{4} = \square$

$\frac{8}{10} - \frac{5}{10} = \square$

$\frac{7}{8} - \frac{5}{8} = \square$

$\frac{3}{5} + \frac{1}{5} = \square$

22. Order the fractions from largest to smallest.

$\frac{1}{3}$ $\frac{1}{5}$ $\frac{1}{10}$ $\frac{1}{2}$ $\frac{1}{8}$ $\frac{1}{4}$ $\frac{1}{11}$ $\frac{1}{7}$

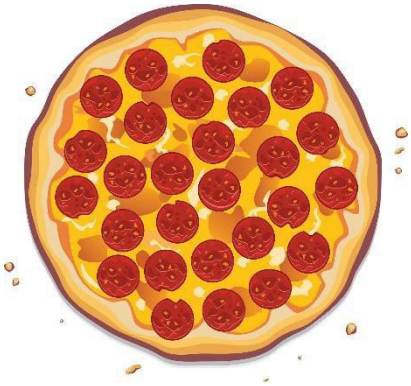
$\square$ $\square$ $\square$ $\square$ $\square$ $\square$ $\square$ $\square$

23. Order the fractions from smallest to largest.

$\frac{7}{9}$ $\frac{1}{9}$ $\frac{5}{9}$ $\frac{6}{9}$ $\frac{4}{9}$ $\frac{2}{9}$ $\frac{8}{9}$ $\frac{3}{9}$

$\square$ $\square$ $\square$ $\square$ $\square$ $\square$ $\square$ $\square$

24. Marley is sharing a pizza with 4 of his friends. How many slices will it take for everyone to get two slices?



Marley needs to cut $\square$ slices for everyone to have two.

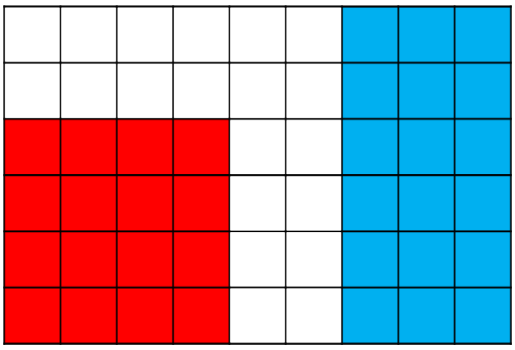
How much of the whole pizza will each child get? Write as a fraction. $\square$

Measurement

25. Use the <, > or = symbols to compare the following.

38 ml		48 ml	273 g		446 g
945 kg		395 kg	300 m		321 m
29 mm		47 mm	362 ml		263 ml
70 L		52 L	28 cm		38 cm

26. Each square is 1 cm long and 1 cm tall. Measure the perimeter of the rectangles.



red rectangle

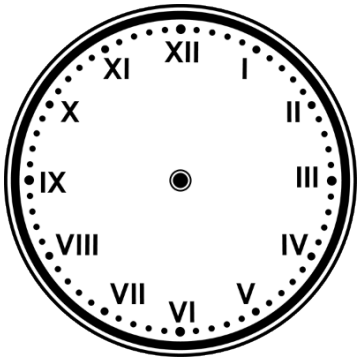
blue rectangle

27. Find the total cost of the items listed:

milk	£1.25
eggs	£1.00
bread	£0.75
cheese	£1.10
apples	£0.80
Total:	

How much change would you receive from a ten pound note?

28. Draw hands on the clocks to show these times.



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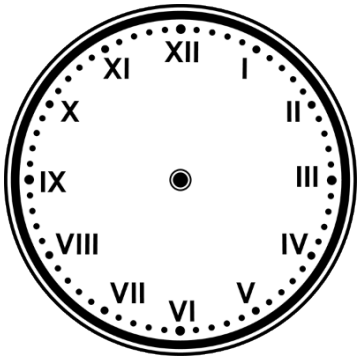

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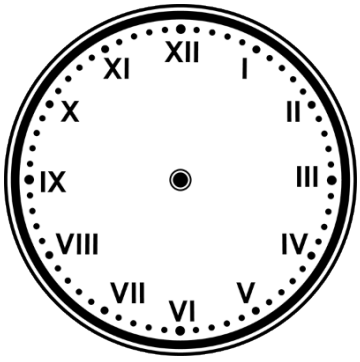


15:57

29. How many seconds are in one minute? ☐

How many days are in the following months?

March	☐	July	☐	February	☐
October	☐	April	☐	November	☐

How many days are in a year? ☐

How many days are in a leap year? ☐

30. Use the <, > or = symbols to compare the following lengths of time.

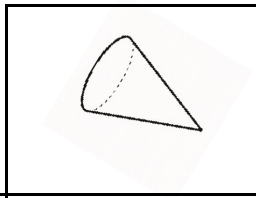
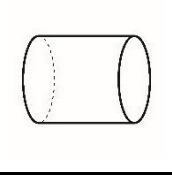
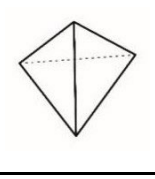
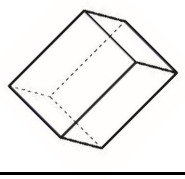
2:00pm – 4:30pm	☐	6:30pm – 7:30pm
9:00am – 3:00pm	☐	8:00am – 2:00pm
4:30am – 7:30am	☐	4:00pm – 7:00pm
4:15pm – 6:30pm	☐	1:00am – 3:30am

Geometry

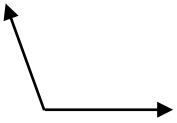
31. Draw the following 2D shapes below their names and mark on their internal angles.

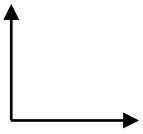
square	triangle	circle	rectangle	pentagon	hexagon

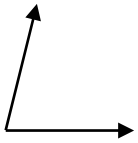
32. Fill in the table with the properties of the 3D shapes.

				
Name of shape				
edges				
vertices				
faces				

33. Circle the right angle.

A 

B 

C 

34. How many right angles make a half turn?

How many right angles make a whole turn?

35. Are the following angles greater than or less than a right angle?

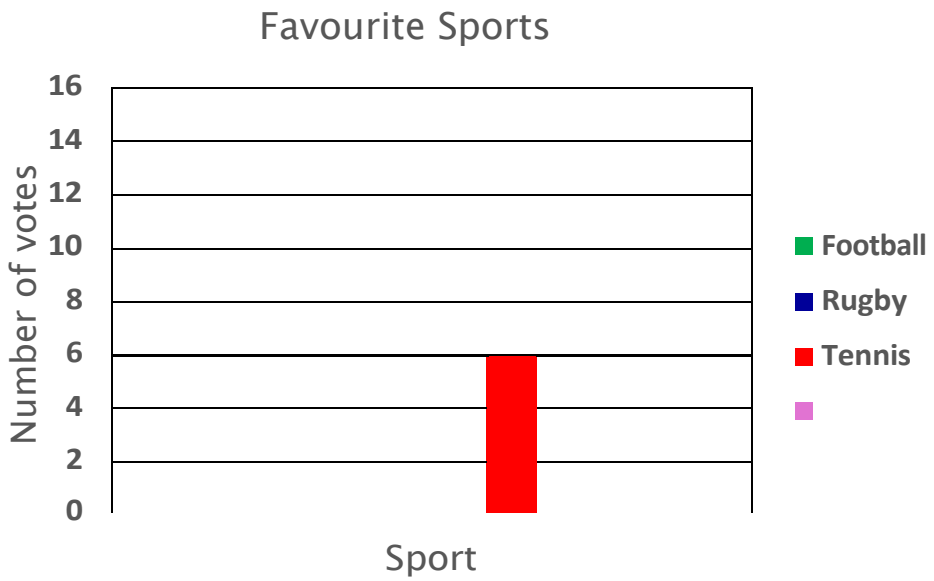
36. Draw a pair of each of the following:

horizontal parallel lines	vertical parallel lines	perpendicular lines

Statistics

37. Draw a bar chart that displays the information from the pictogram.

Sport	Votes (1 ball = 2 votes)
Football	
Rugby	
Tennis	
Basketball	



38. What is the most popular sport?

How many more people like football than tennis?

How many fewer people voted for basketball than rugby?

How many people voted in total?

Baseline Maths Assessment – Year 4

Number and Place Value

1. 16, 20, 24, 28, 32, 36
   32, 40, 48, 56, 64, 72
   200, 250, 300, 350, 400, 450
   400, 500, 600, 700, 800, 900

2. 759 299 925
   693 315 793

3. 2 hundreds 5 tens 6 ones
   3 tens 5 hundreds 2 ones

4. > < <

5. four hundred and twenty-six seven hundred and eight
   six hundred and ninety-five eight hundred and thirty-seven

Addition and Subtraction

6. 846 219
   378 208

7. 239 743
   382 661

8. 927 420
   170 938

9. 1,126

10. 326

11. Estimate around 400 (500 + 400 = 900); actual answer is 407

12. 446
    895

Multiplication and Division

13.

18	72	28	12	40
32	4	88	36	6
9	36	9	44	96

14. £2 x 6 = £12

15. £12 x 4 = £48

16. 4 bags of sweets

Fractions

17.

$\frac{5}{10}$ $\frac{6}{10}$ $\frac{7}{10}$ $\frac{8}{10}$

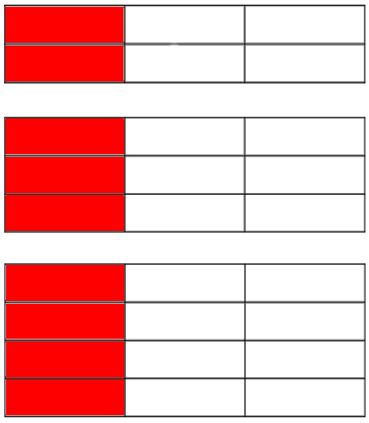
$\frac{5}{10}$ $\frac{4}{10}$ $\frac{3}{10}$ $\frac{2}{10}$

18. red $\frac{1}{7}$ blue $\frac{6}{7}$

19. 4 8

6 14

20.



$\frac{2}{6}$

$\frac{3}{9}$

$\frac{4}{12}$

21. $\frac{3}{4}$ $\frac{3}{10}$

$\frac{2}{8}$ $\frac{4}{5}$

22. $\frac{1}{2}$ $\frac{1}{3}$ $\frac{1}{4}$ $\frac{1}{5}$ $\frac{1}{7}$ $\frac{1}{8}$ $\frac{1}{10}$ $\frac{1}{11}$

23. $\frac{1}{9}$ $\frac{2}{9}$ $\frac{3}{9}$ $\frac{4}{9}$ $\frac{5}{9}$ $\frac{6}{9}$ $\frac{7}{9}$ $\frac{8}{9}$

24. 10 slices $\frac{2}{10}$ or $\frac{1}{5}$

Measurement

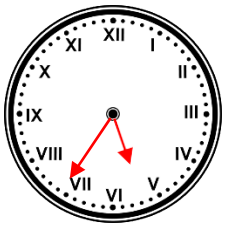
25.

38 ml	<	48 ml	273 g	<	446 g
945 kg	>	395 kg	300 m	<	321 m
29 mm	<	47 mm	362 ml	>	263 ml
70 L	>	52 L	28 cm	<	38 cm

26. red rectangle: 16 cm blue rectangle: 18 cm

27. £4.90 total £5.10 change

28.



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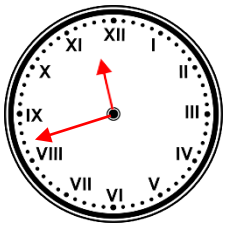

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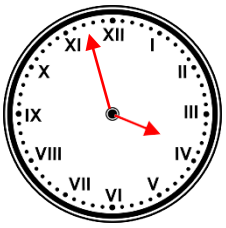


15:57

29. 60 seconds

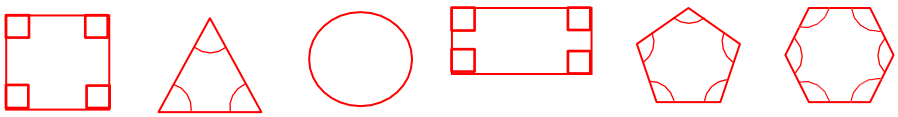
31	31	28 (also 29 in a leap year)
31	30	30

365 days in a year
366 days in a leap year

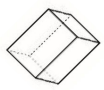
30.

2:00pm – 4:30pm	>	6:30pm – 7:30pm
9:00am – 3:00pm	=	8:00am – 2:00pm
4:30am – 7:30am	=	4:00pm – 7:00pm
4:15pm – 6:30pm	<	1:00am – 3:30am

31.



32.

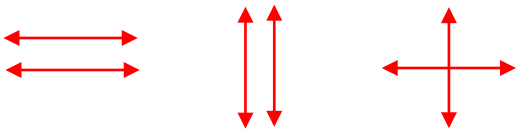
				
Name of shape	cone	cylinder	pyramid	cuboid
edges	1	2	6	12
vertices	1	0	4	8
faces	2	3	4	6

33. B

34. 2 right angles
4 right angles

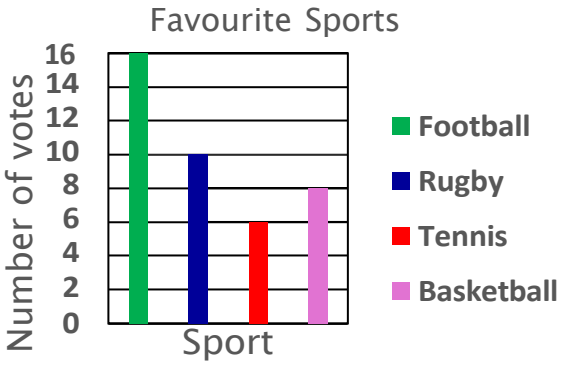
35. greater than less than

36.



Statistics

37.



38. football
10 more
2 fewer
40 people in total